Process Development and Troubleshooting in Microfabrication

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Background and Experience

- 3+ years of cleanroom experience
- End-to-end development of microfluidic devices via photolithography and soft lithography
- Designed devices for controlled electrochemical experiments
- Troubleshooting via root cause analysis and metrology
- Achieved repeatable channel fabrication across 10+ photolithography devices and 100+ soft lithography devices

Agenda:

- Photolithography experience + technical challenge
- Soft lithography experience + technical challenge

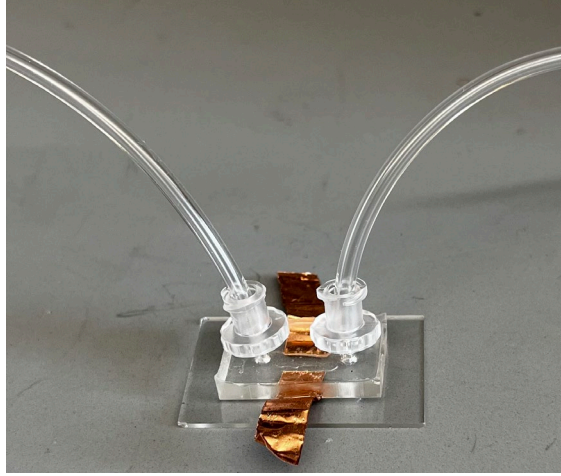
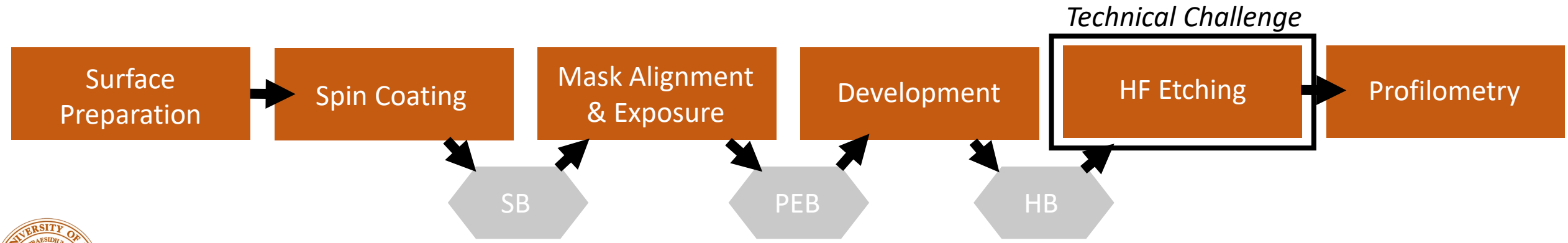


Fig 1: Soft lithography microfluidic device

Microfluidic cell via Photolithography

Fig 2: Schematic of deposit growth in the channel between Cu electrodes

- Designed microfluidic cell for electro-extraction of Cu from dilute electrolyte
- Device to facilitate 2D planar deposition
- Glass substrate to enable in-situ optical visualization of deposition patterns
- Channel depth of 10 μm to approximate quasi-2D transport conditions



Spin coating

- Uniform SU8 coating using vacuum chuck spin coater
- Control Parameters
  - Thickness target - 25 μm ± 3 μm

  $t = kp^2/w^{1/2}$, where k = spinner constant, typically 80-100

  p = resist solids content in percent

  w = spinner rotational speed in rpm/1000
  - Speed of spincoater (rpm) helps control film thickness
  - Time
- Failure modes:
  - Edge bead → poor mask contact
  - Thickness non-uniformity → exposure variation
  - Striations → flow instability
- Soft bake removes solvent and stabilizes film

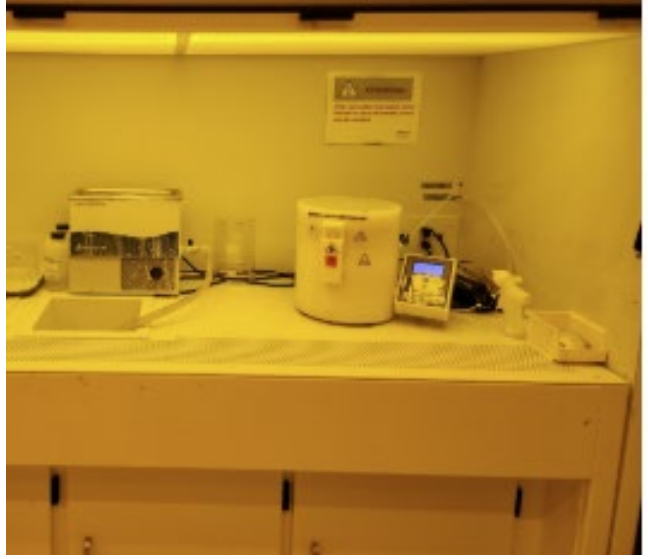


Fig 3: Laurell Spincoater

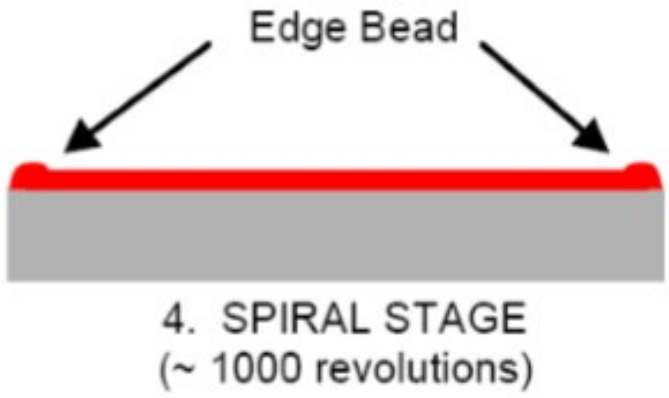


Fig 4: Schematic of Edge bead

Mask Alignment & Exposure

- Negative photoresist – SU8
- Control Parameters
  - Dose (Nominal: 240 mJ/cm^2, process window:$\pm$ 10–15%)
  - Contact mode (soft)
  - Mask quality
- Failure Modes
  - Underexposure
  - Overexposure
  - Poor contact
- Process adapted to account for optical transmission differences in glass substrates

Fig 5: SUSS MA6 Mask Aligner

Development and Post Exposure Bake

- Substrate immersed in SU8 developer
- Development is followed by an IPA rinse and dried with N_2
- Control Parameters:
  - Temperature ramp
  - Time
- Failure modes
  - Incomplete crosslinking
  - Film Distortion due to overbake

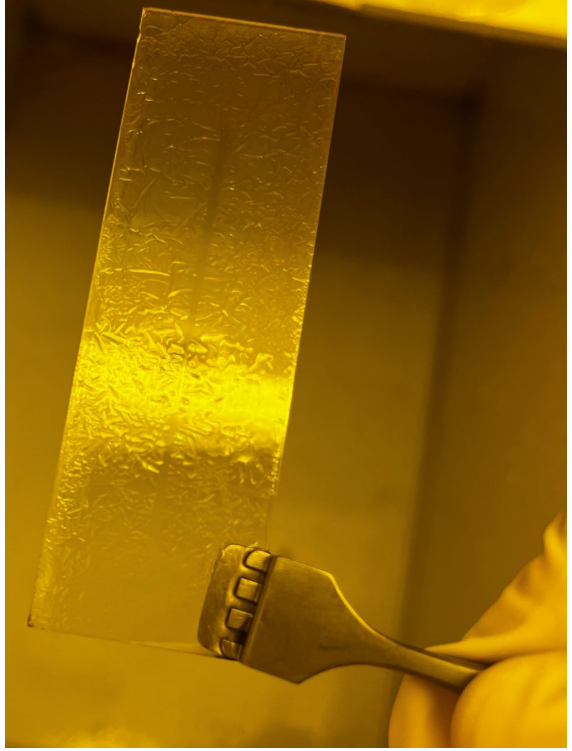


Fig 6: Film distortion post development and exposure bake

HF etching

- Used 6% HF solution to etch exposed glass regions
- Control parameters:
  - HF concentration
  - Etch time
  - Temperature (25°C)
  - SU8 adhesion and crosslinking
- Failure modes:
  - Widened channels due to undercut
  - Uncontrolled etch due to mask delamination
  - Non-uniform etch depth across substrate
- Etch rate ≈ 10 μm / 17 min (~0.6 μm/min)
- Etch stopped via DI water rinse and drying
- Channel depth validated using profilometer measurements

Fig 7: HF Hood

Troubleshooting SU-8 delamination during HF etch

- SU8 lifting caused widened channels, poor edge definition and loss of repeatability
- Eliminate delamination and achieve repeatable channel formation <2 μm feature deviation and stable masking
- Hypotheses
  - Adhesion failure
  - Incomplete crosslinking
  - Limited mask durability in HF
- Need for DOE-style analysis
- Identify dominant parameters controlling mask survival during etching

Experiments to isolate the parameter driving failure mode

- Responses
  - Delamination (Pass/Fail)
  - Channel depth verified by profilometer
  - Feature integrity via optical inspection to detect distortion

Factor	Low	High
Dehydration bake	10 min	20 min
Exposure dose	220 mJ/cm²	260 mJ/cm²
Hard bake temp	120°C	150°C

Dektak 6M Stylus Profilometer

Zeiss Axioscope 2 MAT Optical Microscope

Fig 8: Tools for inspection

Results

- Results indicated adhesion had the strongest correlation with mask survival
- Samples with increased dehydration bake showed reduced delamination
- Increasing the exposure energy from the nominal value resulted in improved cross-linking

Microfluidic cell via soft lithography

- Rapid prototyping of PDMS based device to study effect of flow parameters on extraction efficiency
- Fabricated device by casting PDMS onto a 3D-printed mold for rapid design iteration
- Maintained channel depth of 10 μm via SLA 3D printing and using the appropriate resin and 3D printer of 1 μm resolution

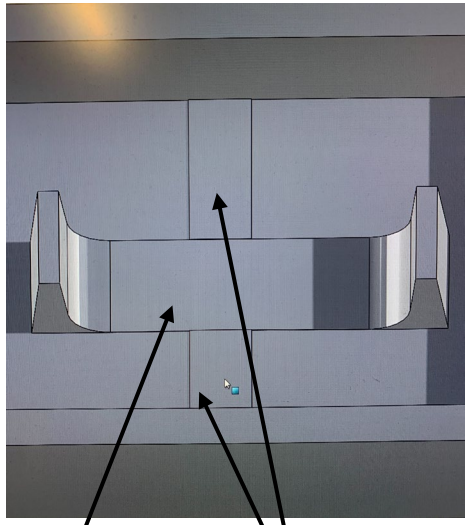


Fig 8: Top view of PDMS mold

Geometry design in CAD → 3D printing of mold → Cast PDMS & degassed → PDMS curing → PDMS bonding to glass → Validated via leak testing

Technical Challenge

Plasma bonding of glass and PDMS

- Operated and optimized process conditions on March Plasma Etcher
- Tuned key parameters to control surface activation and bonding quality
  - RF power (105 W)
  - Gas concentration (O_2, 30%)
  - Chamber pressure
  - Etch time
- Created SOP for plasma bonding to produce microfluidic device

Fig 9: March RIE Plasma Etcher

Troubleshooting plasma bonding

- PDMS–glass bonding failed after 60 s O_2 plasma exposure
- Observed poor adhesion and leakage in devices
- Hypothesis: prolonged plasma exposure modified the surface, reducing effective bonding sites (Si–OH groups)
- Systematically reduced exposure time (60 s $\rightarrow$ 20 s)
- Evaluated bonding quality across conditions
- Achieved consistent, strong bonding at optimized exposure time of 22 ± 2 s
- Eliminated leakage and improved process reliability

Fig 10: Plasma bonded device

Translating my experience to wafer fabrication

- Process Optimization
  - Apply DOE to define process windows for lithography and etch steps
- Yield Improvement
  - Use SPC (X-MR, I-MR charts) to monitor process stability and detect drift
  - Reduce wafer-to-wafer and run-to-run variability
- Troubleshooting
  - Structured root cause analysis
  - Distinguish between tool-induced vs process-induced failures